

Market-aware Feature Engineering

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1 Methodology

The 60-feature observation $o_t^{(60)} \in \mathbb{R}^{60}$ used by the team’s PPO baseline is a point-statistic snapshot: per-tick means, last- and last-three-tick views, and episode-historical means of bid, cost, and win-rate signals. It does not encode three classes of information that are recoverable in closed form from the per-tick history cache the simulator already maintains: (i) *time derivatives* of those statistics, (ii) *higher-order moments* of the within-tick competitor-bid distribution, and (iii) *opponent-aggression proxies* that ratio-encode auction outcomes. Since the simulator’s 47 competitors are unobserved, every non-Markov signal the policy might use must be surfaced into o_t explicitly; this extension adds such features at the observation layer while leaving the action space, transition model, reward, and PPO objective untouched.

We define the market-aware observation

$$o_t = [o_t^{(60)}; \phi_t^{(1)}; \phi_t^{(2)}] \in \mathbb{R}^{98}, \quad (1)$$

where $\phi_t^{(1)} \in \mathbb{R}^{20}$ (*Tier 1*) selects 20 named statistics that the simulator’s history cache already maintains per tick but that $o_t^{(60)}$ does not surface (per-slot win counts and exposure rates, raw conversion counts, CPA-exceedence rates, and episode cumulatives), and $\phi_t^{(2)} \in \mathbb{R}^{18}$ (*Tier 2*) is a set of features derived in closed form from the same cache via three primitives: a length- W least-squares slope slope_W , a length- W sample standard deviation std_W , and an episode-history z -score $z_t(x) = (x_t - \bar{x}_{1:t}) / \text{std}_{1:t}(x)$, all with $W=5$. The 18 Tier 2 features are organised into six categories that target the three blind spots above: competitor-bid distribution shape, opponent-aggression proxies, ad-slot dynamics, regime z -scores, signed momentum of win rate / ROI / conversion rate, and a budget-pacing residual; their definitions are listed in the Appendix.

The augmented observation is paired with the same clipped-PPO surrogate, the same actor and critic MLP architectures (apart from the input layer width, which grows from 60 to 98), the same 20-period rollout schedule on training periods 7–26, and the same 10M-step budget as the $o_t^{(60)}$ baseline. Because the modification is confined to the observation layer, this extension is by construction orthogonal to those that act on the reward (Lagrangian shaping), on rollout sampling (score-weighted reweighting), or on the policy class (GRU temporal encoder, LLM policy), and could in principle be composed with any of them.

2 Results

We evaluate the trained policy on the held-out period 27 using the 48-advertiser sweep protocol (one deterministic rollout per advertiser at that advertiser’s true (`budget`, `CPAConstraint`)). The market-aware policy attains a mean evaluation score of **24.85 ± 2.09** (s.e.m., $n=48$), reported in Table ?? alongside the 16- and 60-feature PPO baselines from the team’s summary table.

Table 1: Held-out period-27 advertiser-sweep evaluation. Baseline numbers are from the team-wide extension comparison; the market-aware run uses the same PPO hyperparameters and 10M-step budget.

Observation	Dim.	Sweep score (mean ± s.e.m.)
PPO baseline (obs.16)	16	24.8
+ 60-feature state	60	27.0
+ Market-aware ($o_t^{(98)}$)	98	24.85 ± 2.09

The market-aware observation is statistically indistinguishable from the 16-feature PPO baseline on this evaluation and underperforms the 60-feature configuration by roughly one s.e.m. The training-time signal on the 20 training periods, however, is healthy: the per-rollout score logged during training rises from ≈ 18 at 1M steps to ≈ 25.4 averaged over the final 1M steps, with `cost_over_budget` ≈ 0.93 (slight under-pacing) and `target_cpa_over_cpa` ≈ 1.05 (slightly inside the CPA target). The training trajectory plateaus around 3M steps and remains flat through 10M, suggesting that the additional 38 features did not introduce instability but also did not unlock further gains beyond what the 60-feature observation already captures on the training distribution.

The most parsimonious explanation for the train/test gap is a *generalisation* failure: a wider observation enlarges the input layer of the actor and critic MLPs and provides more degrees of freedom along which the policy can specialise to the training periods 7–26. Several Tier 2 features (windowed slopes, z -scores, ratios) are also high-variance scalar transforms of the same underlying signals already present in $o_t^{(60)}$, which can amplify estimation noise without adding information that survives the train/test distribution shift to period 27. Two natural follow-ups are (a) a permutation-importance pass on the trained model to identify which Tier 2 features actually carry signal in the training distribution, followed by a leaner o_t that retains only those keys, and (b) regularisation aimed at the wider input layer (e.g. input dropout or an L2 penalty on the first-layer weights of the actor and critic). Both are deferred to future work; the present extension should be read as evidence that simply expanding the observation vector along plausible market-regime axes is not sufficient on this benchmark.

3 Appendix

A.1 Windowed primitives

For any tracked scalar signal $x_{1:t}$ and window length W , with $\bar{k} = (W-1)/2$ and $\bar{x} = \frac{1}{W} \sum_{k=0}^{W-1} x_{t-W+1+k}$,

$$\text{slope}_W(x_t) = \frac{\sum_{k=0}^{W-1} (k - \bar{k})(x_{t-W+1+k} - \bar{x})}{\sum_{k=0}^{W-1} (k - \bar{k})^2}, \quad \text{std}_W(x_t) = \sqrt{\frac{1}{W} \sum_{k=0}^{W-1} (x_{t-W+1+k} - \bar{x})^2}. \quad (2)$$

The slope is the closed-form OLS estimate against an integer time index; the episode z -score $z_t(x) = (x_t - \bar{x}_{1:t})/\text{std}_{1:t}(x)$ uses cumulative mean and std over the episode prefix, with $\text{std}_{1:t}$

floored at a small ε to guard against constant prefixes.

A.2 Tier 1 surfaced statistics (20 keys)

Per-slot win counts $\{\text{bid_success_count_slot_}j\}_{j=1,2,3}$ in last/last-three/historical views; per-slot exposure rates $\{\text{exposure_mean_slot_}j\}_{j=1,2,3}$ in the same three views; last-three and historical raw conversion counts; last/last-three/historical CPA-exceedence rates; and the episode cumulatives Y_t, C_t, CPA_t . These quantities are already maintained by the simulator’s per-tick history update, so Tier 1 is implemented as a selection-only change to the JSON observation key list with no code change to the environment.

A.3 Tier 2 derived features (18 keys)

Below lwc_t denotes the within-tick mean of the least winning competitor cost (the auction floor faced by the agent), b_t the agent’s mean submitted bid at tick t , p_t the mean predicted conversion probability over the tick’s impression batch, wr_t the win rate, and C_t, B, T_{left} the cumulative spend, total budget, and remaining time fraction.

- **Competitor distribution shape (7).** $\text{slope}_5(\text{lwc})$ and $\text{std}_5(\text{lwc})$ capture trend and volatility of the bid floor; the coefficient of variation $\text{std}_5/\text{mean}_5$ separates absolute from relative volatility; $z_t(\text{lwc})$ locates the current tick within the episode’s own scale; the right-tail width $\text{lwc}_t^{(01\%)} - \text{lwc}_t^{(10\%)}$ and ratio $\text{lwc}_t^{(\text{max})}/\text{lwc}_t^{(\text{mean})}$ expose “whale” bidders whose presence the baseline mean smooths out; slope_5 of the percentile gap tracks whether the tail is widening.
- **Opponent-aggression proxies (2).** Because competitor bids are unobservable, we infer opponent stance from auction outcomes: $\alpha_t^{\text{opp}} = \text{lwc}_t/p_t$ (price the market clears per unit of predicted conversion) and the agent’s own competitiveness $\beta_t = b_t/\text{lwc}_t$.
- **Slot dynamics (3).** The cost ratio $c_t^{(1)}/c_t^{(3)}$ between the top and floor ad slot, the analogous exposure ratio, and the slope_5 of slot concentration. Together they signal whether premium ad real estate is being contested or vacated.
- **Regime z -scores (2).** z_t of pv-volume intensity and of win rate, locating the current tick within the episode’s empirical distribution rather than against an absolute scale.
- **Momentum / acceleration (3).** slope_5 of wr_t , of the ROI proxy p_t/lwc_t , and of the per-tick conversion rate.
- **Budget-pacing residual (1).** $\rho_t = C_t/B - (1 - T_{\text{left}})$, with $\rho_t > 0$ indicating cumulative spend ahead of elapsed time and $\rho_t < 0$ indicating underpacing.